

Winter term 2025/26

Image Acquisition and Analysis in Neuroscience

Assignment Sheet 2

Solution has to be uploaded by November 10, 2025, 10:00 a.m., via eCampus

If you have questions concerning the exercises, please use the forum on eCampus.

- Please work on this exercise in **small groups** of 3 students. Submit each solution only once, but clearly indicate who contributed to it by forming a team in eCampus. Remember that all team members have to be able to explain all answers.
- Please submit your answers in PDF format, and your scripts as *.py/*.ipynb files. If you are using [Jupyter notebook](#), please also export your scripts and results as PDF.
- Admission to the exam requires **at least 75 points** (50% of the regular points from all six sheets) and presenting an answer in class **at least once**.

Exercise 1 (Equilibrium Magnetization, 8+1 Bonus Points)

Chapter 3, Slide 19 discusses how the equilibrium magnetization on which Magnetic Resonance Imaging is based arises from the alignment of nuclear magnetic moments with an external magnetic field $\mathbf{B}_0$, and from a tiny excess of spins in the lower energy state. It also claims that the magnitude M_0 of the resulting macroscopic magnetization is proportional to the strength B_0 of the external field. This exercise will help you better understand how these statements fit together.

- a) Given the overall number n of hydrogen nuclei in a voxel, and the Boltzmann statistics $r = n_{\downarrow}/n_{\uparrow}$, i.e., the ratio of spins with antiparallel $\downarrow$ (higher energy) to parallel $\uparrow$ (lower energy) alignment, compute the number of spins in the lower energy state that are *not* canceled by another spin in the higher energy state. Explain why this leads to the following equation for the strength of the resulting macroscopic magnetization:

$$M_0 = \frac{1 - r}{1 + r} \frac{n \hbar \gamma}{2} \quad (1)$$

Plug in values for a clinical scanner $B_0 = 1.5 \text{ T}$ and body temperature $T \approx 310 \text{ K}$ as well as an approximate value $n \approx 6.69 \times 10^{19}$ of protons in one cubic millimeter of water to obtain a numerical result, including physical units. (4P)

You can earn one bonus point for researching the order of magnitude of the magnetic moment of a compass needle for comparison (please cite your source).

- b) The claimed proportionality between M_0 and B_0 holds approximately, for values of B_0 that are small enough to lead to a Boltzmann statistic that is very close to one. For notational simplicity, we could express its dependence on B_0 as

$$r = \frac{n_{\downarrow}}{n_{\uparrow}} = e^{\zeta B_0} \quad \text{with} \quad \zeta = -\frac{\hbar \gamma}{k_B T}. \quad (2)$$

Compute the derivative of Equation (1) with respect to B_0 (2P). Use it to specify a Taylor approximation of $M_0(B_0)$ around $B_0 = 0$ that is truncated after the linear part. (1P) Observe that it is linear in B_0 and confirm that, for clinical field strengths such as $B_0 = 1 \text{ T}$ or $B_0 = 3 \text{ T}$, it yields very similar numerical values as the exact Equation (1). (1P)

Exercise 2 (Spatial Encoding in MRI, 6 Points)

- The MRI isocenter is the point in the scanner where the gradient magnetic fields change directions. In other words, the field strength at the isocenter equals the main magnet's B_0 at all times. Assuming $B_0 = 3\text{ T}$ and gradient strength $G_z = 40\text{ mT/m}$, compute the center frequency f_c and bandwidth Δf of an RF pulse (both in Hz) that excites a 2 mm thick slice around the isocenter. How can we excite a slice of the same width, but away from the isocenter? (3P)
- The explanations that were given in the lecture assume that the number of times we sample the MR echo equals the number of columns in the k -space matrix. In practice, most modern MR scanners oversample the signal. For example, for a 256×256 image, an oversampling factor of two would take 512 complex measurements per echo, and reduce them to 256 complex numbers via a low-pass filter. Explain what type of artifact can be avoided in this way. Does this remove artifacts in the frequency encoding direction, the phase encoding direction, or both? (3P)

Exercise 3 (Challenges in MR Image Reconstruction, 11+5 Bonus Points)

In theory, MR image reconstruction should be as simple as taking a 2D inverse Fourier transformation. In this exercise, you will meet a few challenges that arise due to imperfections in the measurements.

- The file `mrimage1d.txt`¹ contains raw measurements from a conventional MR scan of dimension 256×256 . They are stored as ASCII text. Each of the 65,536 lines contains two numbers, the real and imaginary components of one measurement. Use those values to fill a complex 256×256 k -space matrix and reconstruct the corresponding magnitude image. Account for the fact that the 2D FFT treats position $(0,0)$ as the origin, while we expect the k -space to be centered, and would also like to visualize the center of our 2D signal in the center of the image. This requires proper use of the `ifftshift` and `fftshift` that were introduced on the previous assignment sheet. Your result should look as in Figure 1. (4P)

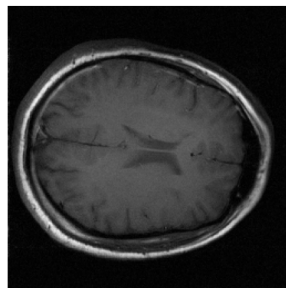


Figure 1: Reconstructed conventional MR magnitude image.

- Visualize magnitudes in the k -space from a). Apply the logarithm to make sure that not only the largest values are visible; properly handle values that are close to zero. Is the spectrum centered as expected? Visualize the real and imaginary parts of the reconstructed image separately. Is the reconstructed image real-valued, up to small effects from random noise? Use the properties of the Fourier Transformation to make a connection between your observations in the spatial and the frequency domains. (4P)
- The file `sepi.npy` contains a vector with 4096 complex numbers, providing the raw MR measurements from a 64×64 EPI experiment. Reconstruct the image that corresponds to these raw measurements. Note that, unlike in the conventional scan you dealt with previously, you now

¹Source: <https://www.cis.rit.edu/htbooks/mri/chap-10/detail/det-h-1.html>

need to account for the zig-zag fashion in which EPI traverses k -space, illustrated in Figure 2 left, by reversing the order, and taking the complex conjugate, of every other row. At this point, your result is expected to look as in Figure 3 left. (3P)

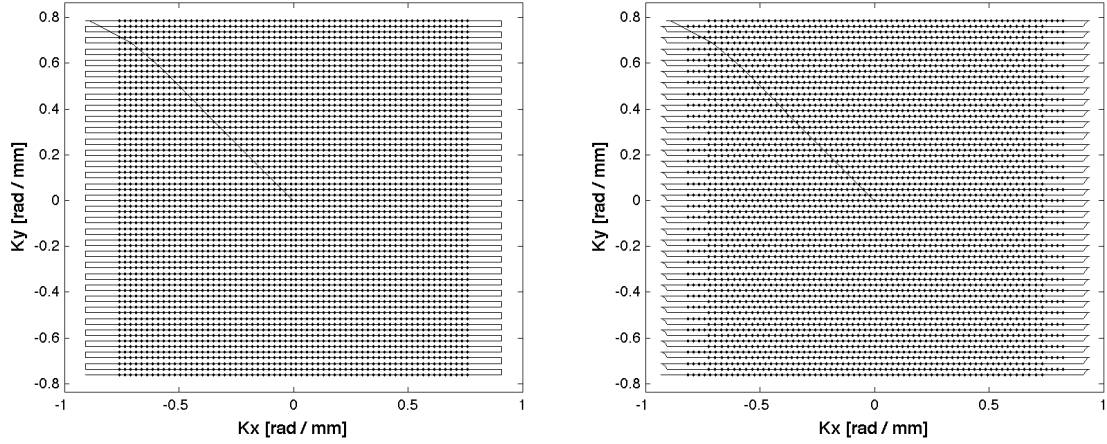


Figure 2: Left: Ideal EPI k -space traversal. Right: EPI k -space traversal with time shift.

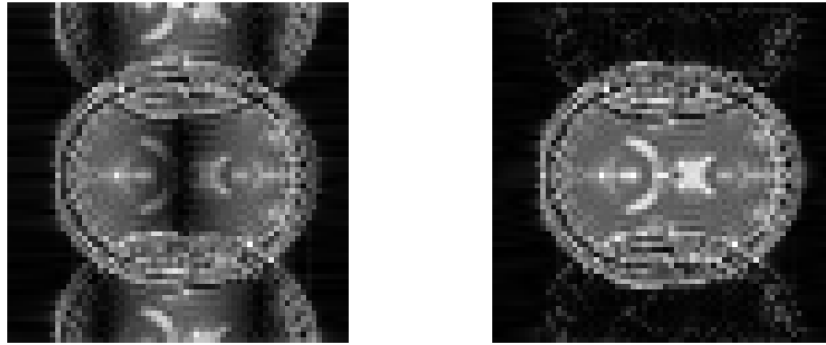


Figure 3: Left: Reconstructed MR image with strong eddy current EPI artifact. Right: Artifact greatly reduced via k -space interpolation

- d) **Bonus:** The image reconstructed in c) still suffers from severe artifacts. They are due to a slight time shift between even and odd echoes, which stems from eddy currents and is illustrated in Figure 2 right. The file `spc.npy` contains 128 complex numbers, which are the raw measurements from two echoes with positive and negative polarity, respectively, and no phase encoding. You can determine the time shift by plotting these two signals and comparing the location of their maxima. You can obtain five bonus points if you manage to reduce the artifact by compensating for the shift via 1D interpolation. Feel free to use interpolation routines from `scipy`. You might also need to adjust the phases to achieve a real Fourier coefficient at the center of k -space. Your final result might look as in Figure 3 right.

Good Luck!